Week 7

Course Level Objectives:

- Explain the role of probability in statistics. (CLO 4)
- Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. (CLO 5)
- Describe and compute confidence intervals. (CLO 6)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Construct a 95% confidence interval as statistic ± 2·SE, given the statistic and its standard error. (CLO 6) **2.** Interpret a confidence interval in context. (CLO 6) **3.** Explain the confidence level as the long-run proportion of intervals that capture the parameter. (CLO 4, CLO 6) **4.** Identify common misinterpretations of a confidence interval. (CLO 6)	e-Text: 3.2 Understanding and Interpreting Confidence Intervals Slides: 3.2 Understanding and Interpreting Confidence Intervals	Participation: Sampling Distribution (Kahoot!) – Mon: review of 3.1 Participation: Confidence Intervals – which of 20 simulated CIs capture μ; fix three misinterpretations; Your Turn: Do These CIs Capture p?	Online Homework: 3.2 Understanding and Interpreting Confidence Intervals (due Sun Oct 11, 11:59 PM) Take-Home Quiz 4 (due Mon Oct 12, 11:59 PM) – covers 3.1–3.3 (all objectives) Test 2 (Fri Oct 30 – Sun Nov 1, proctored) – covers P.1, 3.1–4.3 (all objectives) Test 2 Study Guide (all objectives) Practice Test 2 (optional; due Thu Oct 29) (all objectives)	WileyPLUS Blackboard Kahoot! Respondus
5. Describe how a bootstrap sample and a bootstrap distribution are created. (CLO 5, CLO 6) **6.** Explain why a bootstrap distribution is centered at the sample statistic. (CLO 5) **7.** Use StatKey to generate a bootstrap distribution. (CLO 5, CLO 6) **8.** Estimate the standard error as the standard deviation of the bootstrap distribution. (CLO 5, CLO 6) **9.** Construct and interpret a 95% bootstrap confidence interval using the standard-error method. (CLO 6)	e-Text: 3.3 Constructing Bootstrap Confidence Intervals Slides: 3.3 Constructing Bootstrap Confidence Intervals (embedded StatKey simulator) StatKey (lock5stat.com/statkey) – Bootstrap Confidence Intervals	Participation: Bootstrap Confidence Intervals – Wed: Your Turn: Bootstrap a Proportion (32 of 50 prefer Brand A): build the bootstrap distribution, read off SE, form the CI	Online Homework: 3.3 Constructing Bootstrap Confidence Intervals (due Sun Oct 11, 11:59 PM) Take-Home Quiz 4 (due Mon Oct 12, 11:59 PM) – covers 3.1–3.3 (all objectives) Test 2 (Fri Oct 30 – Sun Nov 1, proctored) – covers P.1, 3.1–4.3 (all objectives) Test 2 Study Guide (all objectives) Practice Test 2 (optional; due Thu Oct 29) (all objectives)	WileyPLUS Blackboard StatKey Respondus

Also this week: Test 1 Retake – Mon Oct 5 – Sun Oct 11 (optional; higher score counts)